Manufacturing Flowchart of Aluminium 2618-T6 Piston

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Material Selection

The piston head requires a material that can withstand high thermal loads, mechanical stress, and continuous impact forces inside a combustion engine. Aluminum 2618-T6 was selected because:

- It has excellent high-temperature strength, preventing deformation during engine operation.
- It offers low density, reducing reciprocating mass and improving engine efficiency.
- Its fatigue resistance helps the piston survive repeated combustion cycles.
- It provides good machinability, allowing accurate production of the thin walls and small internal features.

Together, these properties make 2618-T6 aluminum ideal for high-performance piston applications.

Stock

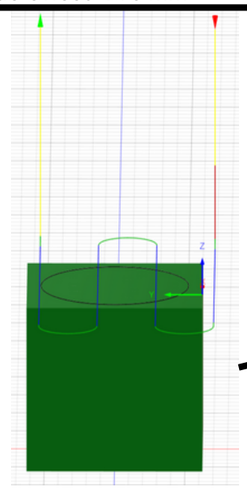
The stock is a cylinder with diameter 12.49 mm and height 13 mm. This provides enough material for machining while avoiding unnecessary waste.

- Diameter: 12.49 mm
- Height: 13 mm
- Stock Top Offset: 1.5 mm
- Stock Side Offset: 1.75 mm

Manufacturing Evaluation

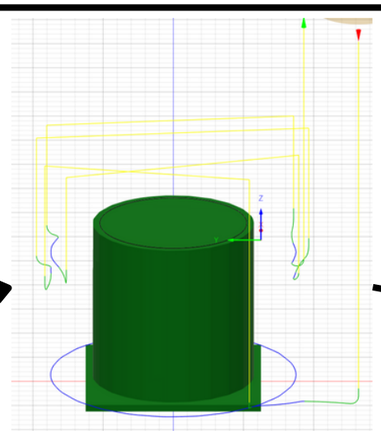
- Number of Operations: 7
- Number of Tools: 4
- Tools Used: T2 (Ø6 mm flat), T3 (Ø1 mm flat), T4 (Ø2.5 mm drill), T6 (Ø0.1 mm ball end)
- Maximum Feedrate: 1000 mm/min
- Maximum Spindle Speed: 5000 rpm
- Cutting Distance: 4061.04 mm
- Rapid Distance: 6279.32 mm
- Estimated Cycle Time: 9 min 13 s

The piston head needed several milling and drilling steps to create the internal cavity and the pin hole correctly. Even though the part is small, it required choosing the right tools, especially because the walls are thin and some cuts are deep. Using computer-aided manufacturing helped keep all dimensions within tolerance, and each toolpath helped improve the quality of the final shape.
Overall, the piston head was made successfully, but adding a final polishing pass could make the surface even smoother.



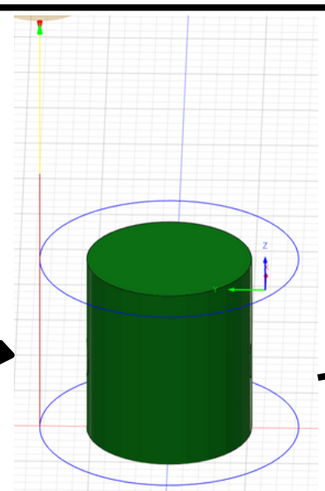
Operation 1/7
Description: Face3
Strategy: Facing
WCS: #0
Tolerance: 0.01mm
Maximum stepover: 4.2mm

Maximum Z: 15mm
Minimum Z: -1mm
Maximum Spindle Speed: 5000rpm
Maximum Feedrate: 1000mm/min
Cutting Distance: 115.41mm
Rapid Distance: 25.4mm
Estimated Cycle Time: 8s (1.4%)
Coolant: Flood



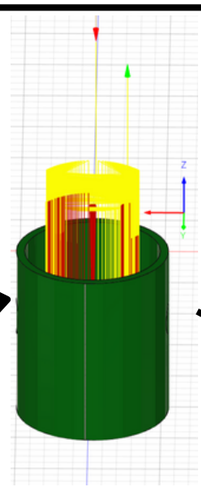
Operation 2/7
Description: Adaptive1
Strategy: Adaptive
WCS: #0
Tolerance: 0.1mm
Stock to Leave: 0.5mm
Maximum stepdown: 13.5mm
Optimal load: 2.4mm
Load deviation: 0.24mm

Maximum Z: 15mm
Minimum Z: -12.5mm
Maximum Spindle Speed: 5000rpm
Maximum Feedrate: 1000mm/min
Cutting Distance: 103.21mm
Rapid Distance: 186.25mm
Estimated Cycle Time: 8s (1.5%)
Coolant: Flood



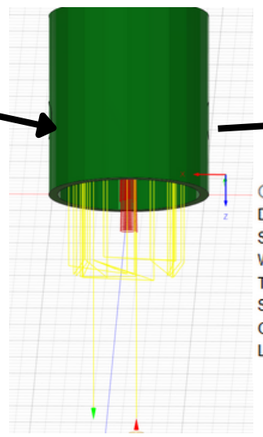
Operation 3/7
Description: 2D Contour2
Strategy: Contour 2D
WCS: #0
Tolerance: 0.01mm
Stock to Leave: 0mm
Maximum stepover: 5.7mm

Maximum Z: 15mm
Minimum Z: -13mm
Maximum Spindle Speed: 5000rpm
Maximum Feedrate: 1000mm/min
Cutting Distance: 127.61mm
Rapid Distance: 64mm
Estimated Cycle Time: 11s (2%)
Coolant: Flood



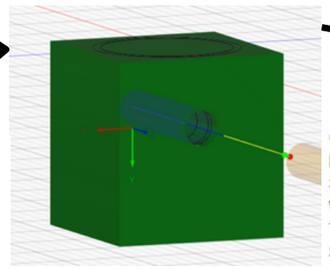
Operation 4/7
Description: Adaptive2
Strategy: Adaptive
WCS: #0
Tolerance: 0.1mm
Stock to Leave: 0mm
Maximum stepdown: 2.5mm
Optimal load: 0.4mm
Load deviation: 0.04mm

Maximum Z: 15mm
Minimum Z: -10mm
Maximum Spindle Speed: 5000rpm
Maximum Feedrate: 1000mm/min
Cutting Distance: 1704.2mm
Rapid Distance: 384.33mm
Estimated Cycle Time: 3m:1s (32.7%)
Coolant: Flood



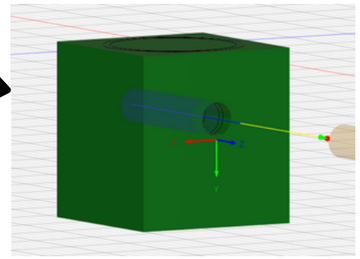
Operation 5/7
Description: 2D Adaptive2
Strategy: Adaptive 2D
WCS: #0
Tolerance: 0.01mm
Stock to Leave: 0.5mm
Optimal load: 0.04mm
Load deviation: 0mm

Maximum Z: 15mm
Minimum Z: -9.5mm
Maximum Spindle Speed: 5000rpm
Maximum Feedrate: 1000mm/min
Cutting Distance: 1978.19mm
Rapid Distance: 5541.8mm
Estimated Cycle Time: 4m:42s (50.9%)
Coolant: Flood



Operation 6/7
Description: Drill2
Strategy: Drilling
WCS: #0
Tolerance: 0.01mm
Stock to Leave: 0mm

Maximum Z: 27.49mm
Minimum Z: 0mm
Maximum Spindle Speed: 5000rpm
Maximum Feedrate: 1000mm/min
Cutting Distance: 16.49mm
Rapid Distance: 38.49mm
Estimated Cycle Time: 1s (0.3%)
Coolant: Flood



Operation 7/7
Description: Drill2 (2)
Strategy: Drilling
WCS: #0
Tolerance: 0.01mm
Stock to Leave: 0mm

Maximum Z: 15mm
Minimum Z: -12.49mm
Maximum Spindle Speed: 5000rpm
Maximum Feedrate: 1000mm/min
Cutting Distance: 15.93mm
Rapid Distance: 39.05mm
Estimated Cycle Time: 1s (0.3%)
Coolant: Flood

In the final two steps, drilling was performed from both sides of the part because Fusion 360 did not generate a through-hole in a single operation.